

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT (2 Questions)

Course Code:	ITA05	Course Title:	COMPUTER VISION
CO Assessed:	CO2 – Manipulation (BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	20 (2 Questions × 10 Marks)
CO5 Statement:	<i>Apply image enhancement and segmentation techniques for extracting meaningful regions from images.(BL3)</i>		
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Section / Batch:	Slot c/ Hostel/2024	Date:	

Code of Conduct

I Y.Thanushma(192425355) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Y.Thanushma

Instructions

- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus Area	Q. Nos.	Marks
Frequency Domain Image Enhancement	Periodic noise removal, Fourier Transform, notch filtering, quality inspection	1	10
Spatial Filtering & Edge-Preserving Enhancement	Bilateral filtering, edge preservation, autonomous vehicle navigation	2	10
GRAND TOTAL:			20 Marks

Annexure A – Scenario Based Assessment

Scenario 1: Manufacturing Quality Inspection

An automated quality inspection system captures images of metal components. Electrical interference introduces repetitive horizontal stripes across the images, making defect detection unreliable.

- Interpret the scenario and identify the type of noise present.
- Explain how this degradation affects inspection accuracy.
- Recommend an appropriate enhancement technique to remove the interference.
- Justify why your selected approach is more suitable than spatial-domain methods.
- Explain your answer in a logical sequence.

Scenario 2: Autonomous Vehicle Navigation

A self-driving car captures road images during heavy rain. Although noise is present, the lane markings and road edges must remain sharp for safe navigation.

- Interpret the scenario and explain the challenge involved.
- Identify the importance of preserving edges while removing noise.
- Recommend a suitable filtering technique.
- Justify why your selected filter preserves important image details.
- Present your explanation clearly and systematically

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding ; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	

Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand		
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Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Question 1: Manufacturing Quality Inspection

a) Scenario Analysis and Noise Identification

The automated quality inspection system captures images of metal components for defect detection. However, electrical interference introduces repetitive horizontal stripes across the images, making it difficult to identify defects accurately. This type of degradation is known as Periodic Noise. It is caused by electrical or electromagnetic interference and appears as repetitive patterns in the image. In the frequency domain, periodic noise is represented by distinct frequency peaks, making it easier to identify and remove.

b) Impact of Periodic Noise on Quality Inspection

Periodic noise significantly reduces the quality of captured images. It hides small defects such as cracks, scratches, and dents on metal components, leading to inaccurate inspection results. As a result, the automated system may incorrectly classify defective products as good or reject good products unnecessarily. This reduces inspection accuracy, increases production costs, and affects overall product quality.

c) Recommended Image Enhancement Technique

The most suitable enhancement technique is Frequency Domain Filtering using the Fourier Transform and a Notch Reject Filter.

Working Process:

- Convert the image from the spatial domain to the frequency domain using the Fourier Transform (FFT).
- Identify the frequency components responsible for periodic noise.
- Apply a Notch Reject Filter to suppress only the unwanted frequencies.
- Use the Inverse Fourier Transform (IFFT) to reconstruct the enhanced image with reduced noise.

This method effectively removes repetitive interference while preserving important image information.

d) Justification for Frequency-Domain Filtering

Frequency-domain filtering is more effective than spatial-domain filtering because periodic noise is concentrated at specific frequencies. A notch filter removes only those unwanted frequencies without

affecting the remaining image content. In contrast, spatial filters such as mean or Gaussian filters blur the entire image and cannot selectively remove periodic patterns. Therefore, frequency-domain filtering provides better defect visibility and higher inspection accuracy.

e) Stepwise Image Restoration Process

The complete image enhancement process is as follows:

1. Capture the image of the metal component.
2. Detect repetitive horizontal stripe noise.
3. Apply the Fourier Transform (FFT).
4. Identify periodic noise frequencies.
5. Apply a Notch Reject Filter to remove unwanted frequencies.
6. Perform the Inverse Fourier Transform (IFFT).
7. Obtain a noise-free enhanced image.
8. Detect defects accurately for quality inspection.

Conclusion

In conclusion, the image degradation is caused by periodic noise due to electrical interference, which reduces the accuracy of automated defect detection. Applying the Fourier Transform along with a Notch Reject Filter effectively removes the repetitive noise while preserving important image details. This enhancement improves inspection accuracy, reduces detection errors, and ensures reliable quality control in manufacturing.

Question 2: Autonomous Vehicle Navigation

a) Scenario Interpretation and Problem Statement

A self-driving car captures road images during heavy rain. Rain introduces unwanted noise into the captured images, reducing their clarity. Despite the presence of noise, important features such as lane markings, road boundaries, and obstacles must remain sharp for safe vehicle navigation. Therefore, the challenge is to remove noise while preserving edge information.

b) Significance of Edge Preservation

Edge preservation is essential because edges represent the boundaries of important objects in an image. In autonomous driving, lane markings, road edges, traffic signs, pedestrians, and vehicles are identified using edge information. If these edges become blurred during noise removal, the vehicle may fail to recognize critical road features, increasing the risk of accidents. Therefore, preserving edges while reducing noise ensures reliable object detection and safe navigation.

c) Recommended Edge-Preserving Filter

The most suitable filtering technique is the Bilateral Filter.

The bilateral filter removes noise by considering both:

- Spatial proximity (distance between neighbouring pixels)
- Pixel intensity similarity (difference in brightness)

Unlike traditional smoothing filters, it smooths similar regions while preserving sharp edges, making it highly suitable for autonomous driving applications.

d) Justification for Selecting the Bilateral Filter

The bilateral filter is preferred because it effectively removes noise without blurring important image details. It preserves lane markings, road boundaries, and object edges while maintaining image sharpness. Traditional filters such as the mean or Gaussian filter smooth the entire image, causing edge loss. Since autonomous vehicles rely on accurate edge information for navigation and obstacle detection, the bilateral filter provides superior performance and improves system reliability.

e) Systematic Workflow for Safe Navigation

The image enhancement process for autonomous navigation follows these steps:

1. Capture the road image during heavy rain.
2. Detect the presence of rain-induced noise.
3. Apply the Bilateral Filter.
4. Remove noise while preserving edges.
5. Detect lane markings and road boundaries.
6. Identify vehicles, pedestrians, and obstacles.
7. Use the enhanced image for accurate navigation and safe driving.

Conclusion

In conclusion, autonomous vehicle navigation requires effective noise removal while maintaining sharp edges for accurate lane and obstacle detection. The Bilateral Filter is the most suitable technique because it smooths noise without blurring critical image features. This improves the reliability of road scene analysis, enhances navigation accuracy, and contributes to safer autonomous driving in adverse weather conditions.

